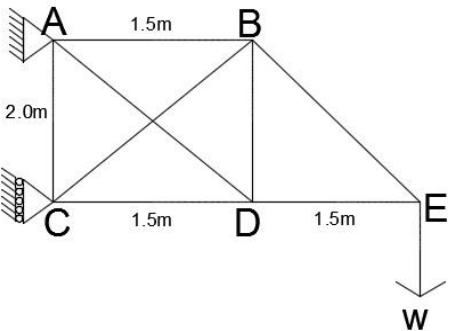
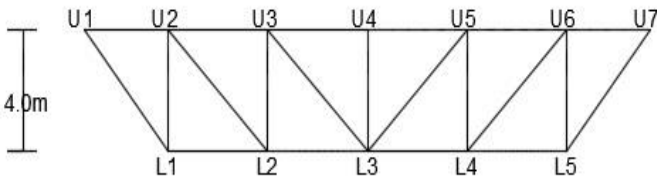
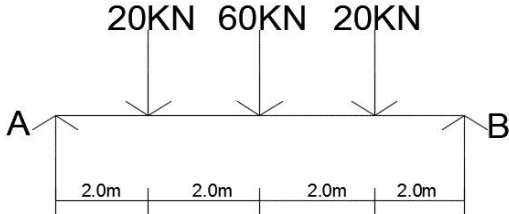
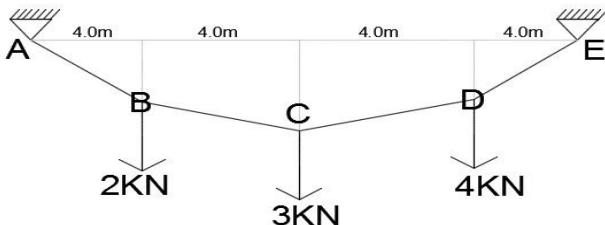
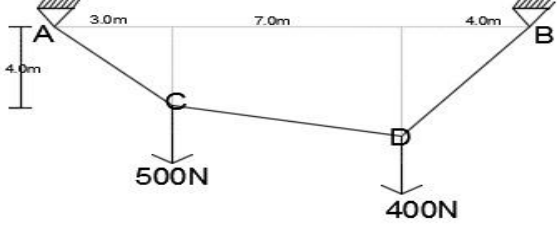
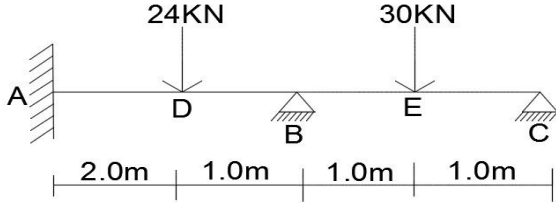
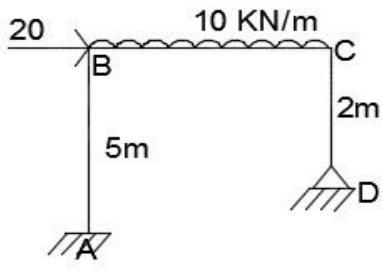
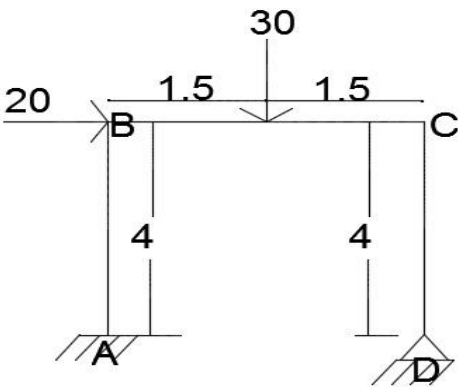
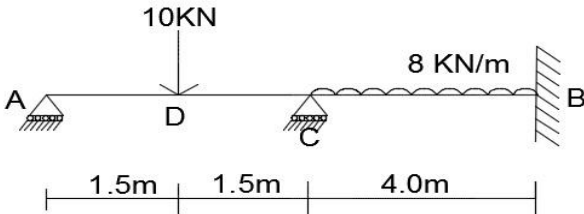
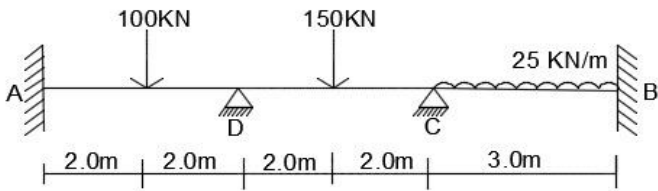
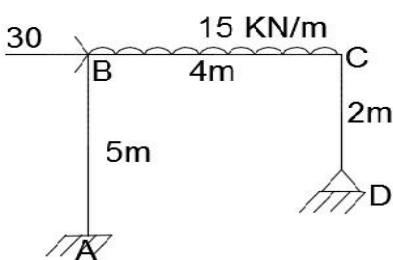


	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : CE Semester :V Subject Code & Name: BTCVC502 Structural Mechanics II Max Marks: 60 Date: Duration: 3 Hr.	
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.	
		(Level/CO)
	Q. 1 Solve Any Two of the following.	CO1
A)	 Analyse the truss frame by energy method .The member are all of the same material	6
B)	 Draw ILD for the various member of the Deck type girder shown in figure.6 Pannels @ 3m each	CO3 6
C)	 A simply supported beam of span 8m loaded as shown in fig.Find the shear force and BM at section 4m from left end .Draw ILD for support reaction,SF,BM.	CO1 6
Q.2	Solve Any Two of the following.	

A)	 Three loads are suspended as shown from the cable ABCDE knowing that $d_c=3\text{m}$ determine A) components of the reaction at E B) the maximum tension in the cable	CO2	6
B)	 A string supported and loaded as shown in fig. Determine the support reaction and maximum tension in segment of the cable	CO2	6
C)	Differentiate lacing and battening	CO1	6
Q. 3	Solve Any Two of the following.		
A)	 By flexibility method analyze the beam and Draw SFD and BMD	CO1	6
B)	 Analyse the frame by Flexibility method	CO1	6

C)	 Analyse the frame by Flexibility method	CO1	6
Q.4	Solve Any Two of the following.		
A)	 Analyse the beam by Stiffness Method	CO1	6
B)	 Analyse the beam by Stiffness Method	CO1	6
C)	 Analyse the beam by Stiffness Method	CO1	6
Q. 5	Solve Any Two of the following.		
A)	State types of elements 1D,2D,and 3D also Plain stress and strain	CO1	6

B)	State the principle of minimum potential energy, also advantage and disadvantages of Finite element method	CO2	6
C)	List the various methods of solving boundries values and properties of stiffness matrix	CO1	6
	*** End ***		